Model question paper

SECOND SEMESTER DIPLOMA EXAMINATION IN INSTRUMENTATION ENGINEERING

BASIC INSTRUMENTATION ENGINEERING

(Maximum marks: 100)

(Time 3 hours)

PART-A

(Maximum marks: 10)

I Answer the following questions in one or two sentences. Each question carries 2 marks.

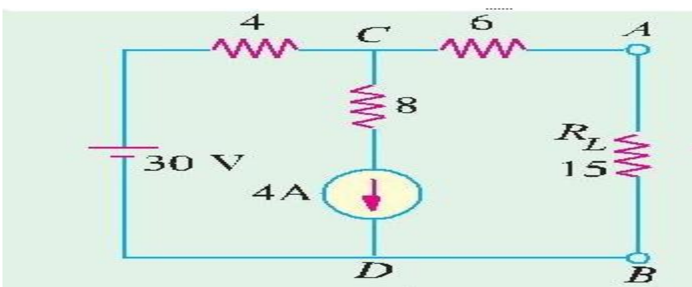
1. Define Linearity
2. State reciprocity theorem
3. Define Early effect
4. List any two advantages of FWR
5. Define pinch off voltage (5x2=10)

PART- B

(Maximum marks: 30)

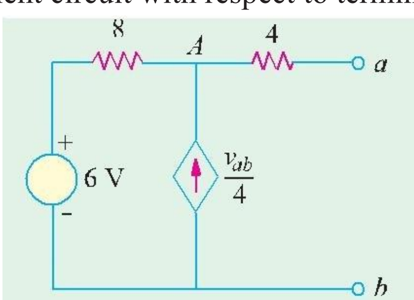
II Answer any *five* of the following. Each question carries 6 marks.

1. Explain the indicating ,recording and controlling functions of instruments
2. State Norton's theorem. Using Norton's theorem, find the constant-current equivalent of the circuit shown in Fig Below.



(a)

3. Explain the working of PNP transistor.
4. Describe the operation of P channel FET
5. Find the Thevenin equivalent circuit with respect to terminals a and b of the network shown in Fig.below



(a)

6. Draw and explain the output characteristics of CE configuration

7. Explain dynamic characteristics of instrumentation system. ` (5x6=30)

PART- C

(Maximum marks: 60)

(Answer one full question from each MODULE. Each question carries 15 marks)

MODULE-I

III Define Error. Explain in detail about different types of errors occurred in measurement systems. (15)

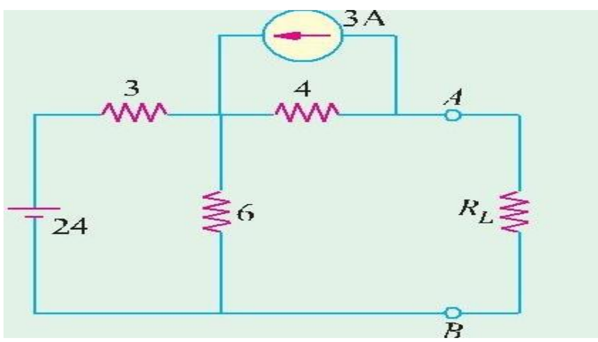
OR

IV (a). Illustrate open loop and closed loop system with examples (7)

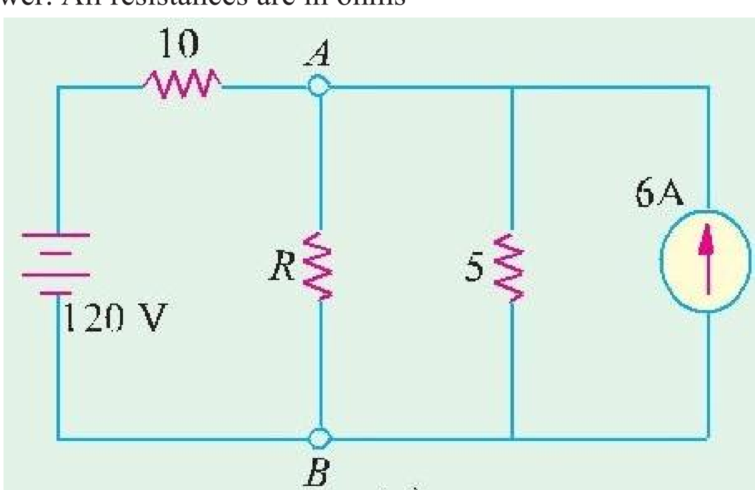
(b). Explain static characteristics of instruments (8)

MODULE-II

V (a) Use Thevenin's theorem, to find the value of load resistance R_L in the circuit of Fig. which results in the production of maximum power in R_L.. All resistances are in ohms (8)



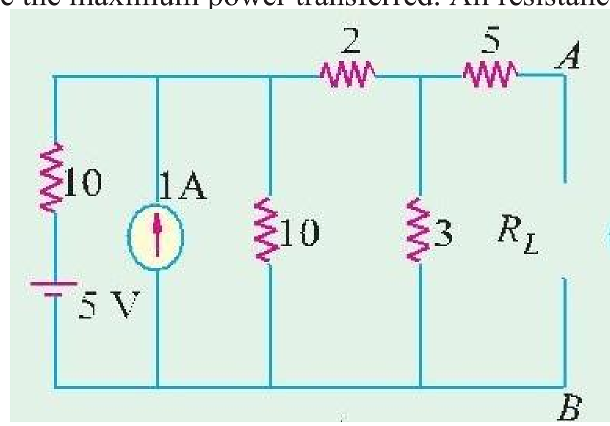
(b) Calculate the value of R which will absorb maximum power from the circuit of Fig..Also, compute the value of maximum power. All resistances are in ohms (7)



OR

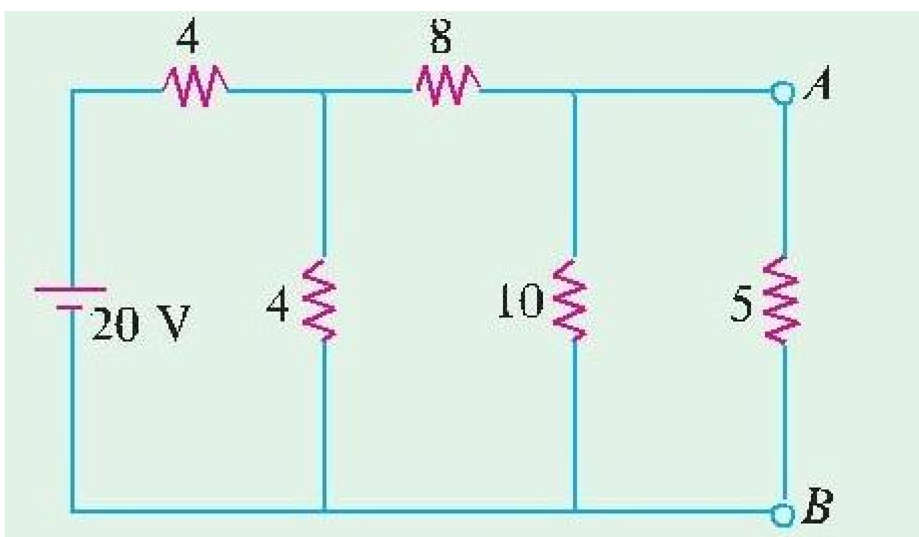
VI

(a) In the circuit shown in Fig. obtain the condition from maximum power transfer to the load R_L. Hence determine the maximum power transferred. All resistances are in ohms (8)



(b)

Apply Norton's theorem to calculate current flowing through 5 – Ω resistor of figure shown below.All resistances are in ohms (7)



MODULE-III

VII (a) With neat sketches, explain the input and out put Characteristics of CB configuration (10)

(b)Explain extrinsic semiconductors (5)

OR

VIII (a) Draw and explain the forward and reverse characteristics of PN junction diode. (10)

(b) Compare CB,CE and CC configuration (5)

MODULE-IV

IX (a) Describe the working of zener diode voltage regulator (8)

(b) Explain the characteristics of P channel FET (7)

OR

X (a) Describe characteristics of N channel MOSFET (8)

(b) Explain the working of DC power supply (7)